

Distributed Volumetric Neural Representation for in situ Visualization and Analysis

Qi Wu^{*†} Joseph Insley^{*†} Victor Mateevitsi^{*} Silvio Rizzi^{*} Kwan-Liu Ma[†]

^{*}Argonne National Laboratory

[†]University of California – Davis

[†]Northern Illinois University

Abstract

Volumetric grids have recently been used by many recent works for representing complex scenes implicitly. A volumetric neural representation can be several orders of magnitude smaller in size while still preserving most of high-frequency details. However, most volumes used in large-scale in situ visualization and analysis are partitioned and generated directly in parallel. Therefore, a compatible technique to create volumetric neural representations for these situations is much needed. In this project, we explore the possibility of constructing and optimizing such a representation for large-scale distributed volumes. We present our preliminary results in this poster. We also outline our plans to integrate our techniques with existing in situ visualization and analysis pipelines.

Volumetric Neural Representation

A volumetric neural representation uses a neural network to directly approximate a volumetric field. Such a neural network takes a spatial coordinate (x, y, z) as the input and outputs a value v corresponding to the volumetric field value at the given spatial coordinate:

$$\Phi: \mathbb{R}^3 \rightarrow \mathbb{R}^D, (x, y, z) \rightarrow \Phi(x, y, z) = v$$

In this work, we focus on scalar field volumes; thus, v would be a single scalar value. To access v encoded by the representation, the neural network needs to be inferred. Our base neural network consists of a multi-resolution hash encoding layer and a small multilayer perceptron network (MLP).

Neural Network Architecture:

To construct a neural representation for distributed volumetric data, we replicate the base neural network on each distributed ranks. Different distributed ranks can then optimize different neural networks locally using different data partitions. The base network is implemented using MPI and GPU accelerated PyTorch.

The base neural network contains a 64×4 MLP and an 8-level hash encoding layer with 4 features per level. The encoding layer uses a hash table of 2^{19} entries with a base resolution 4.

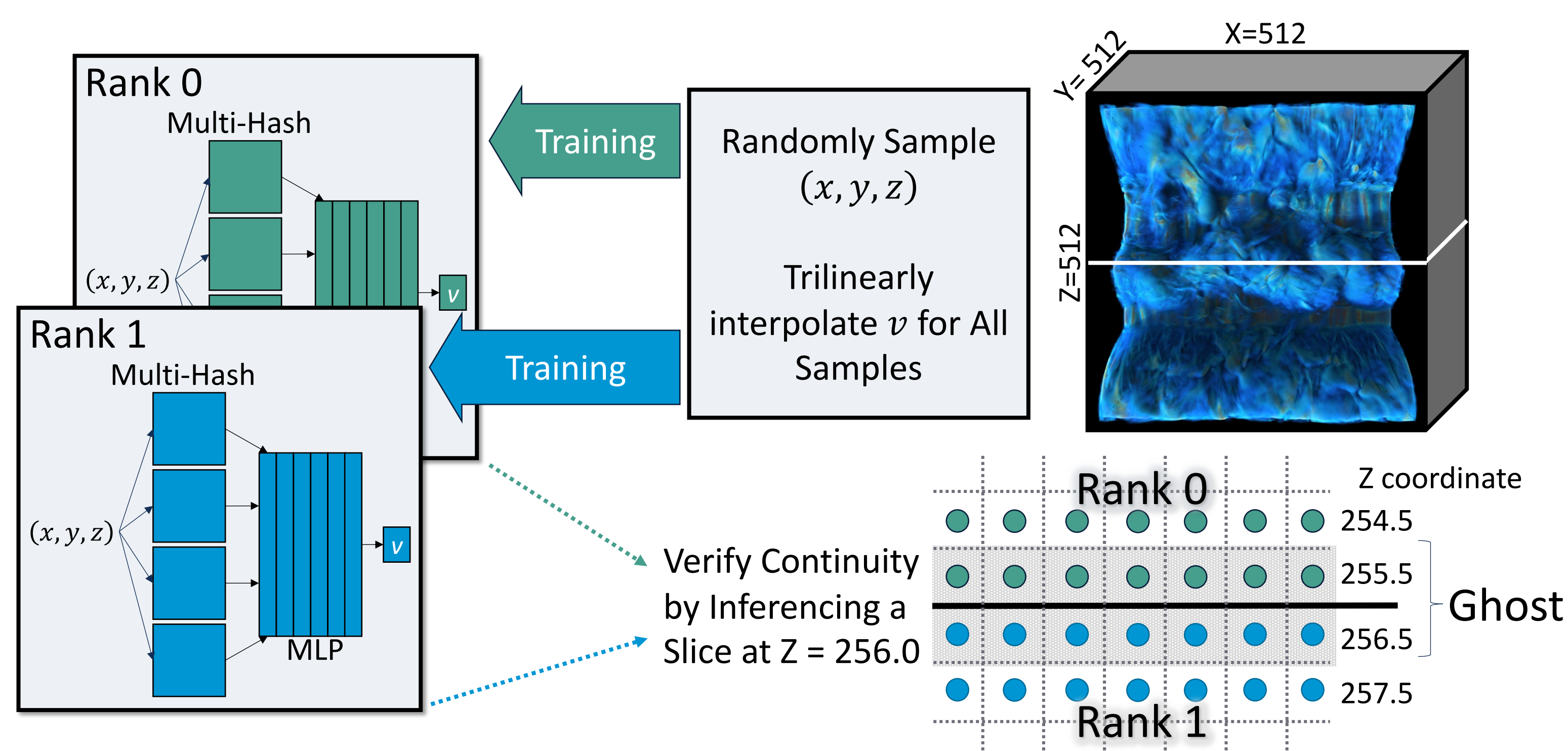


Figure 1. The training and evaluation setup described in this poster. As a proof of concept, we use a regular volume that is pre-partitioned into two. Ghost voxels are also computed during the partitioning process.

Training

The upper part of Fig. 1 describes our network training setup.

We independently run two instances of the base network using MPI. Each network is trained using samples that are randomly selected within its domain.

All the sample values are calculated with trilinear interpolation.

In each training step, exactly 65536 random samples are taken in each individual partition.

Evaluation

The data continuity across the partition boundary should be preserved by our networks. We verify this in three ways:

- We compare slices calculated using both networks on the boundary.
- We calculate the slice accuracy on the boundary for both networks.
- We render the 3D grid reconstructed using both networks. Then, we visually examine the data continuity.

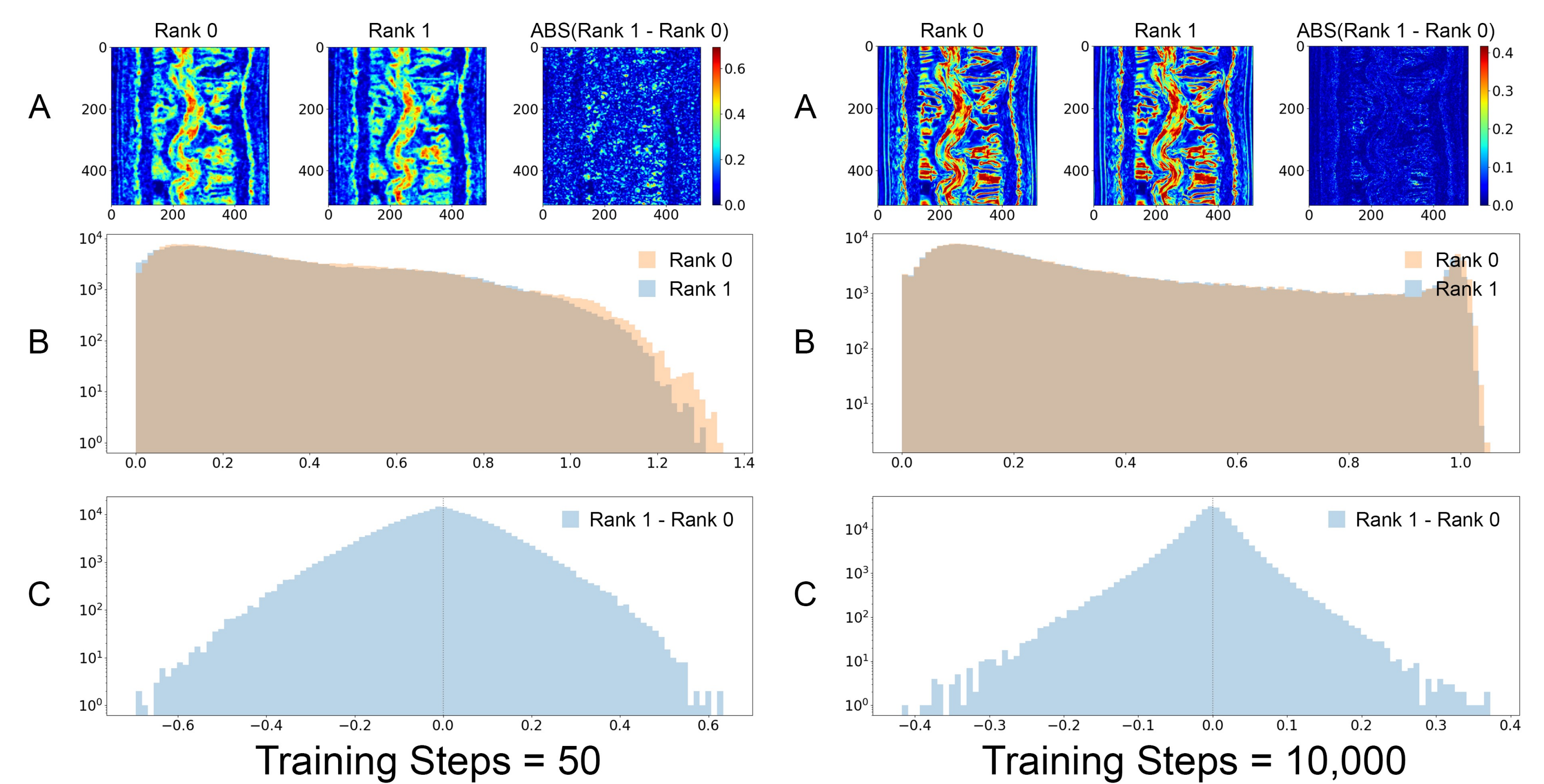


Figure 2. Evaluating the data continuity across the boundary after 50 and 10,000 steps. A) Slices calculated using both networks on the partition boundary. B) The histograms of slices. C) The histogram of the pixel differences between slices.

Results

- Fig. 2 shows the comparison between boundary slices. We can see that there is a discrepancy on the boundary in the early stage of training. However, the discrepancy diminishes as the training continues.

- The PSNR of both boundary slices with respect to the ground truth at different training steps are listed here:

Training Steps	1	100	1,000	10,000
Rank 0	16.2dB	18.0dB	23.1dB	27.6dB
Rank 1	16.1dB	18.1dB	23.1dB	27.5dB

- Fig. 3 shows that the reconstruction accuracy is low in the early stage of training. However, partitioning the volume does not lead to additional visual artifacts.

Our results indicate that our technique is very likely reliable.

Next Steps: Large Scale Data

- Search for the best network setup via hyperparameter tuning. Then, study how the best setup changes as the problem scales up.
- Integrate distributed neural representation with existing in situ infrastructure (e.g., ECP Alpine) and allow it to be accessed via VTKm and Numpy Array APIs.

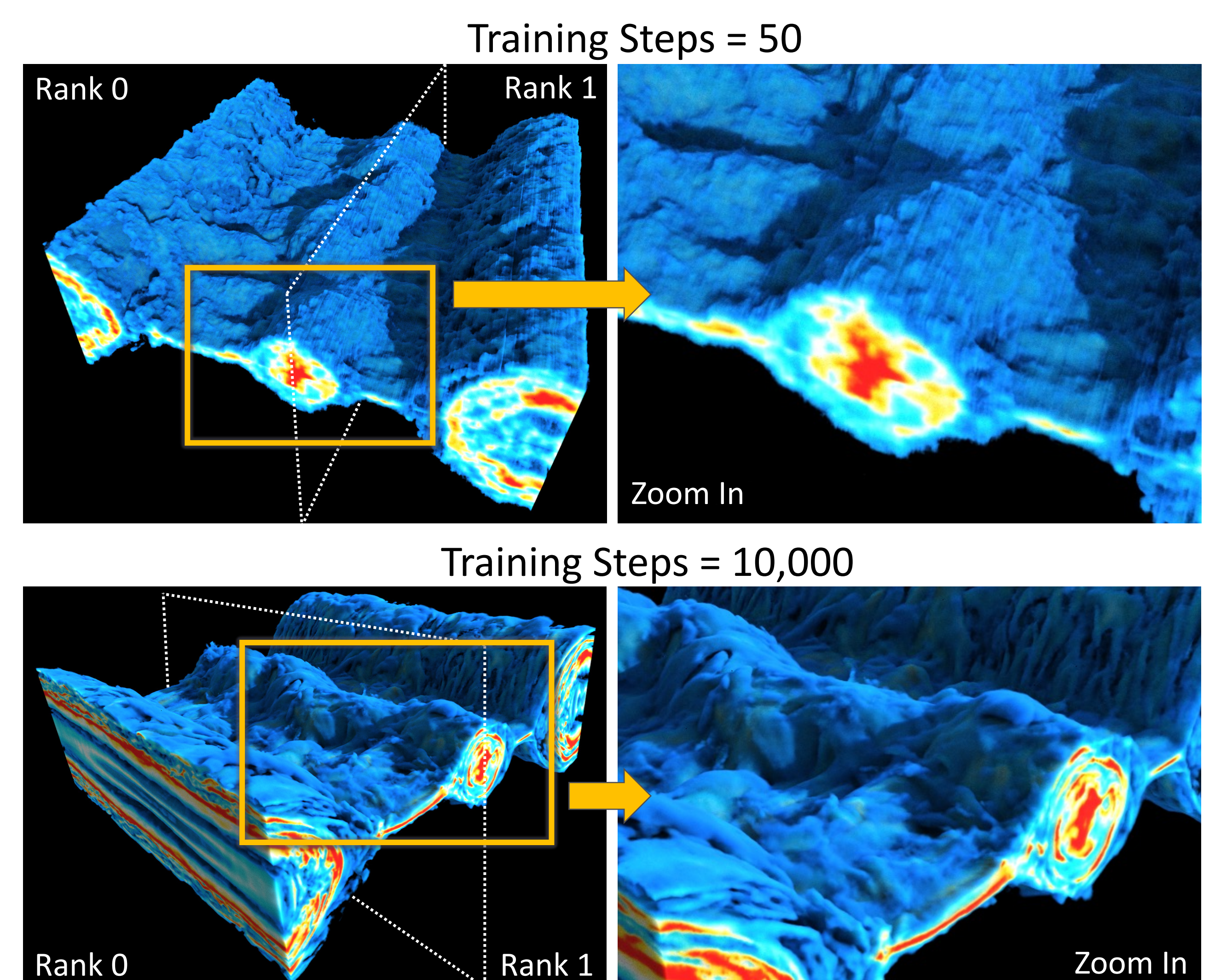


Figure 3. Comparing the rendering results of the reconstructed grid after 50 (top) and 10,000 (bottom) training steps. A zoom-in view on the boundary is also provided for each case. The magnetic reconnection data [1] is used.

[1] Guo, Fan, et al. "Formation of hard power laws in the energetic particle spectra resulting from relativistic magnetic reconnection." *Physical Review Letters* 113.15 (2014): 155005.

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